**Project Report:**

# Enhancing Telemedicine Efficiency in Bangladesh: A Decade of DUTP's Experience

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# Enhancing Telemedicine Efficiency in Bangladesh: A Decade of DUTP's Experience

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**Abstract**

Since the 1920s, telemedicine has evolved through various phases and taken different forms worldwide. The Dhaka University Telemedicine Programme (DUTP) has been serving rural communities in Bangladesh for the past ten years. While it started with a center-based model, the program has continuously adapted, navigating various socio-economic challenges along the way. This project aims to provide a concise summary of DUTP, highlight its achievements, and identify areas for improvement—both for the software development team and the organizational team. However, fluctuations in patient visits and software engagement raise important questions about the effectiveness of different organizational models and the usability of key features. Lastly, by analyzing vitals and basic health data from 35,000 patients across nearly 200 centers, I attempted to uncover key health trends, particularly focusing on hypertension in rural communities.

The primary research objectives include: (1) Identifying when, where, and under what organizational model patient numbers were high or low; (2) Assessing the usage and operator comfort levels with the telemedicine software; (3) Predicting hypertension based on BMI category, age, gender, smoking habit, asthma, and diabetes using machine learning models.

As DUTP is using a software which can be accessed at the link: `https://telemedbd.net` and maintaining a MySQL database for service delivery, I used the database for analysis. Since data was collected primarily for the purpose of running the software properly, processing it initially posed challenges. The study progresses with data preprocessing and feature engineering using Python libraries. Machine learning techniques were also employed for predictive modeling. Initially, Logistic Regression was applied to identify key predictors of hypertension but yielded low accuracy. To improve performance, Random Forest, Gradient Boosting, and XGBoost were explored. A Combined Approach (SMOTE + Downsampling) was implemented to handle class imbalance, ensuring more robust predictions. F1-score and ROC-AUC were used for model evaluation.

Key findings reveal significant fluctuations in patient visits across different centers, influenced by time periods, operator efficiency, and community awareness and locality. Software usage analysis highlighted potential areas for

operator experience improvements. In predictive modeling, ensemble methods outperformed logistic regression, identifying age, BMI category, and diabetes as the most influential predictors of hypertension. This prediction can help our doctors get an idea of the patient's condition before creating a prescription.

This research contributes to the optimization of telemedicine services by identifying bottlenecks in patient engagement and enhancing predictive analytics for chronic disease management. The insights can inform organizational structure, policy decisions, software enhancements, and resource allocation in future telemedicine initiatives.

# 1 Introduction

## Background

Lack of organizational effectiveness, staff motivation, patient satisfaction, and trustworthiness are found to be the main barriers to telemedicine expansion in Bangladesh [4]. Despite these, we have been running this telemedicine project with a philanthropic and research-oriented approach. After a decade, we faced the necessity of a formal quantitative method to analyze our progress, weaknesses, and strengths. A comprehensive analysis and evaluation of the telemedicine program provide evidence-based recommendations for improving patient engagement, software usability, chronic disease management, and long-term program success.

## Literature Review

It has been shown that there are only 3.6 doctors for 10,000 people [4]. For this reason, 70% of people consult village doctors with little or no formal training for curative purposes [1]. To overcome this situation, DUTP started its journey in 2015 with web-based software, a digital stethoscope, and file upload support system [3, 5]. DUTP has passed through different phases in its journey. In 2019, during the pandemic of Corona, it started its hotline-based consultation service [7].

## Research Questions

This study seeks to address the following key research questions:

- **Patient Engagement & Trends:**
  - When and where was patient volume highest and lowest?
  - How did different organizational models impact patient engagement?
- **Software Usability & Operator Experience:**
  - How frequently were different DUTP web-based software features used?
  - What is the time gap between a visit being created and a prescription being generated?
- **Predictive Analytics for Hypertension:**
  - Can hypertension be accurately predicted using BMI category, age, gender, smoking habit, asthma, and diabetes?
  - Which machine learning model provides the best predictive performance?

## Contribution

This research offers a unique blend of healthcare, data science, and technology, providing actionable insights into telemedicine effectiveness:

- First comprehensive analysis of a decade-long DUTP dataset in Bangladesh.
- Bridges the gap between telemedicine adoption, software usability, and predictive healthcare analytics.
- Implements a novel approach combining machine learning with Stratified K-Fold Cross-Validationand Borderline-SMOTE to enhance hypertension prediction.
- Provides policy recommendations for optimizing DUTP's services and software design.

## Outline of the Research

The research is structured as follows:

- **Introduction** – Provides background on DUTP, research objectives, analysis, and significance.
- **Literature Review** – Explores previous studies on telemedicine adoption, machine learning in healthcare, and software usability.
- **Methodology** – Details data collection, preprocessing, feature engineering, and machine learning models used.
- **Results and Analysis** – Presents findings on patient trends, software usage, and hypertension prediction performance.
- **Discussion and Implications** – Interprets results, highlights key takeaways, and suggests improvements in telemedicine implementation.
- **Conclusion and Future Work** – Summarizes contributions, research limitations, and potential future enhancements.

## 1.1 Problem Statement and Motivation

## research gap

While telemedicine in Bangladesh has been a subject of study, several gaps remain in the existing literature that this research aims to address. A significant area of concern is the lack of in-depth analysis of patient demographics and organizational models that influence telemedicine usage. Previous studies, such as Hossain et al. (2020), focused primarily on the general benefits of telemedicine, without evaluating specific factors like patient numbers across different models and settings [2]. The operational

efficiency of telemedicine software, including its usability by operators, remains underexplored. While Rahman (2021) highlighted the adoption of telemedicine, few studies have assessed the comfort levels of operators and their interaction with telemedicine platforms in Bangladesh [6].

Additionally, predictive healthcare models using telemedicine data are still emerging. Although there has been research on predictive models in healthcare, limited studies have focused on using machine learning to predict chronic diseases, such as hypertension, in telemedicine settings in Bangladesh. Existing models tend to be basic and lack the necessary complexity to predict hypertension accurately, especially with factors like BMI, smoking habits, and comorbid conditions (e.g., asthma, diabetes). Studies by Mollah et al. (2021) and Hasan (2022) primarily discuss telemedicine's role during the pandemic, but there is a scarcity of research that incorporates predictive analytics for disease diagnosis using telemedicine data [**?**, **?**].

Moreover, the issue of data quality and preprocessing in telemedicine systems has not been adequately addressed in the literature. As noted in the current research, data in telemedicine systems, such as the one used by DUTP, is often collected primarily for operational purposes, not for analysis, which complicates data preprocessing and feature extraction. This study aims to fill this gap by focusing on improving data preprocessing techniques, using advanced machine learning models, and handling class imbalance, which has not been sufficiently explored in the Bangladeshi context. By addressing these gaps, this research contributes to a deeper understanding of telemedicine in Bangladesh, with an emphasis on predictive health outcomes and operational improvements within telemedicine systems.

## Motivation

The motivation behind this study stems from the growing need to enhance the efficiency and effectiveness of telemedicine systems in Bangladesh. Telemedicine holds immense potential in addressing healthcare access issues, particularly in rural and underserved areas where there is limited access to medical professionals and facilities. However, despite the rapid adoption of telemedicine platforms, significant gaps persist in understanding the operational dynamics, patient engagement, and predictive capabilities of these systems, especially in the context of chronic disease management.

The specific gap this research aims to address is the lack of a comprehensive analysis of patient usage patterns across different organizational models, the usability of telemedicine software by healthcare operators, and the application of machine learning for predictive healthcare. This challenge highlighted a critical gap in the way telemedicine data is being utilized. The lack of research focusing on predictive health analytics in telemedicine systems in Bangladesh inspired the decision to integrate machine learning techniques, such as Logistic Regression, Random Forest, and XGBoost, to predict hypertension more accurately. By addressing the issue of class imbalance using techniques like SMOTE and Downsampling, this study aims to improve the predictive power of telemedicine systems, ultimately contributing to better disease management, early intervention, and more personalized healthcare.

This research is driven by the belief that improving telemedicine's operational ef-

ficiency and predictive capabilities will not only enhance the quality of remote healthcare in Bangladesh but also provide a model for other developing nations facing similar healthcare challenges. By bridging these gaps, this study aspires to contribute to the global body of knowledge on telemedicine and to strengthen the healthcare ecosystem in Bangladesh.

## 1.2 Research Objective

### Main Objectives

The overarching goal of this research is to evaluate and enhance the efficiency of telemedicine systems in Bangladesh by analyzing patient usage patterns, assessing the operator's frequency of using different features of DUTP's telemedicine software, and developing a predictive model for hypertension using machine learning techniques. This study aims to provide valuable insights into the operational, usability, and predictive aspects of telemedicine, ultimately contributing to improved healthcare delivery.

### Specific Objectives

1. **Analyze Patient Usage Patterns:**
    - Identify the factors that contribute to high or low patient numbers within different organizational models of telemedicine services.
    - Assess the temporal and spatial trends in patient usage to understand peak periods and regional variations in telemedicine consultations.
2. **Evaluate Software Usability:**
    - Assess the usability of healthcare operators using the DUTP's telemedicine software to understand operational challenges and areas for improvement.
    - Identify the training needs and technical barriers faced by operators to ensure smoother integration of telemedicine platforms into healthcare workflows.
3. **Predict Hypertension Using Machine Learning:**
    - Develop a predictive model for hypertension using key variables, including BMI category, age, gender, smoking habit, asthma history, and diabetes.
    - Apply and compare machine learning algorithms, such as Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, to identify the most accurate predictors of hypertension.
4. **Address Data Imbalance Issues:**
    - Implement techniques like SMOTE (Synthetic Minority Over-sampling Technique) and Downsampling to handle class imbalance in the dataset.

- Evaluate the effectiveness of these techniques in improving the predictive performance of hypertension models.

5. **Evaluate Model Performance:**

   - Measure the performance of the predictive models using metrics such as F1-score and ROC-AUC to ensure robust

6. **Evaluate Model Performance:**

   - Measure the performance of the predictive models using metrics such as F1-score and ROC-AUC to ensure robust and reliable predictions.
   - Compare the outcomes of different machine learning models to identify the most efficient approach for hypertension prediction in the telemedicine context.

By achieving these objectives, this research will contribute to improving telemedicine system performance, enhance patient care through predictive analytics, and provide actionable recommendations for both software optimization and healthcare delivery in Bangladesh.

# 2 Methodology

## 2.1 Study Design

This study adopts an "analytical research design" focused on evaluating and enhancing the efficiency of telemedicine systems in Bangladesh. The research will analyze patient usage patterns, assess operator's functionality usage with DUTP's telemedicine software, and develop a predictive machine learning model for hypertension using primary data from the telemedicine software data.

## 2.2 Primary Data

- **Target Population:** The target population consists of patients who have used the doctor's consultation services provided by DUTP. Other official records are also used.
- **Target Variable(s):**
  - Hypertension Diagnosis: A binary variable indicating whether a patient has been diagnosed with hypertension.
  - Software's Functionality Usage: A statistical visualization of the percentages of usage of different functionality of the software.
  - Usage Patterns: Variables related to patient usage, including consultation frequency, time of consultation, and geographic region, with the highest-using population properties when the patient's number was at peak.
- **Sampling Design and Sample Size Estimation:** As I worked with the entire dataset of 40,000 patients, this type of sampling can be referred to as passive sampling or non-probability sampling, specifically convenience sampling or census sampling.
- **Data Collection Methods:** Data will be collected from the telemedicine software database maintained by DUTP, which includes patient demographics, medical history, consultation details, and other relevant health information such as blood pressure, BMI, and diabetes.

## 2.3 Statistical Models/Methods

- **Descriptive Statistics:** Descriptive analysis will be performed on patient demographics and consultation patterns to identify trends and usage characteristics.
- **Predictive Modeling:** Machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, will be employed to predict hypertension based on the selected variables. A Combined Approach using SMOTE and Downsampling will be applied to handle class imbalance.

Performance will be evaluated using metrics such as F1-score, ROC-AUC, and accuracy.

- **Statistical Testing:** Chi-square tests will be used to identify statistically significant differences in usage patterns, operator feedback, and health outcomes.

## 2.4 Ethical Considerations

- **Ethical Approval:** Approval from the Authority of DUTP that their data can be used for all research activities in compliance with ethical guidelines for research involving human subjects.
- **Informed Consent:** Informed consent has been obtained from healthcare operators. Patients will be assured that their participation is voluntary, and confidentiality will be maintained.
- **Data Privacy and Confidentiality:** All patient data will be anonymized to protect personal and medical information. Data will be stored securely and used exclusively for research purposes.

# 3 Data Analysis and Results

## 3.1 Descriptive Statistics Summary

### 3.1.1 Data Overview

- The dataset contains 35,645 records.
- It includes both categorical and numerical variables, covering demographic, health, and lifestyle attributes.

### 3.1.2 Demographic Insights

| Gender | Percentage | Count |
|---|---|---|
| Female | 59.3% | 21,119 |
| Male | 40.5% | 14,442 |

Table 1: Gender Distribution in the Dataset

| Marital Status | Percentage | Count |
|---|---|---|
| Married | 70.2% | 25,036 |
| Unmarried | 26.2% | 9,324 |

Table 2: Marital Status Distribution in the Dataset

| Occupation | Percentage | Count |
|---|---|---|
| Housewife | 44.3% | 15,786 |
| Student | 15.6% | 5,558 |
| Farmer | 13.0% | 4,628 |

Table 3: Occupation Distribution in the Dataset

| Religion | Percentage | Count |
|---|---|---|
| Islam | 91.1% | 32,488 |
| Hindu | 7.8% | 2,776 |

Table 4: Religion Distribution in the Dataset

- **Gender**: The majority are female (21,119), followed by male (14,442).
- **Marital Status**: Married individuals (25,036) dominate, followed by unmarried (9,324).
- **Occupation**: The top professions are Housewife (15,786), Student (5,558), and Farmer (4,628).
- **Religion**: Islam (32,488) is the most common, followed by Hindu (2,776).

### 3.1.3 Health-Related Features

| Health Condition | Percentage | Count |
|---|---|---|
| Smoking Habit | 7.0% | 2,482 |
| Asthma | 3.9% | 1,398 |
| Diabetes | 3.86% | 1,377 |

Table 5: Health-Related Features in the Dataset

- Sometimes medical history is not well detected to the rural people.

### 3.1.4 Age Group Distribution

| Age Group | Count |
|---|---|
| 19-35 | 9,845 |
| 36-50 | 8,301 |
| 51-65 | 7,956 |
| 66+ | 5,349 |

Table 6: Age Group Distribution in the Dataset

- The 19-35 age group is the largest (9,845 individuals), followed by 36-50 (8,301).
- The 66+ age group has the smallest representation (5,349).
- There is a fairly balanced distribution across middle-aged groups (36-50 and 51-65).

### 3.1.5 Patient's Occupation

| Occupation | Percentage |
|---|---|
| Housewife | 44.3% |
| Student | 15.6% |
| Farmer | 13.0% |
| Teacher | ¡0.01% (3 cases) |

Table 7: Occupation Distribution of Patients

### 3.1.6 Blood Pressure Descriptive Statistics Summary

| Statistic | Systolic Blood Pressure (mmHg) |
|---|---|
| Count | 28,915 |
| Mean | 115.17 |
| Standard Deviation (Std) | 20.04 |
| Minimum | 0 (Possible data entry errors) |
| 25th Percentile (Q1) | 100 |
| Median (Q2) | 110 |
| 75th Percentile (Q3) | 120 |
| Maximum | 245 |
| Skewness | 0.057 (Almost symmetric distribution) |
| Kurtosis | 5.60 (Heavier tail in distribution) |

Table 8: Systolic Blood Pressure Descriptive Statistics

| Statistic | Diastolic Blood Pressure (mmHg) |
|---|---|
| Count | 28,781 |
| Mean | 74.84 |
| Standard Deviation (Std) | 13.31 |
| Minimum | 0 (Possible data entry errors) |
| 25th Percentile (Q1) | 70 |
| Median (Q2) | 75 |
| 75th Percentile (Q3) | 80 |
| Maximum | 807 (Extreme outlier, likely an error) |
| Skewness | 11.79 (Highly skewed, indicating extreme outliers) |
| Kurtosis | 632.67 (Extremely high, suggesting extreme values) |

Table 9: Diastolic Blood Pressure Descriptive Statistics

## 3.2 Time-Oriented Data

### 3.2.1 Patients by Month

| Month | Patient Count |
|---|---|
| January | 3,633 |
| February | 3,979 |
| March | 4,187 |
| April | 3,245 |
| May | 2,931 |
| June | 2,903 |
| July | 2,668 |
| August | 2,081 |
| September | 2,529 |
| October | 2,982 |
| November | 3,153 |
| December | 3,318 |

Table 10: Patient Distribution by Month

### 3.2.2 Quarterly Patients

| Quarter | Patient Count |
|---|---|
| Q1 (Jan-Mar) | 11,799 |
| Q2 (Apr-Jun) | 9,079 |
| Q3 (Jul-Sep) | 6,717 |
| Q4 (Oct-Dec) | 8,453 |

Table 11: Quarterly Patient Distribution

### 3.2.3 Yearly Patient Trends

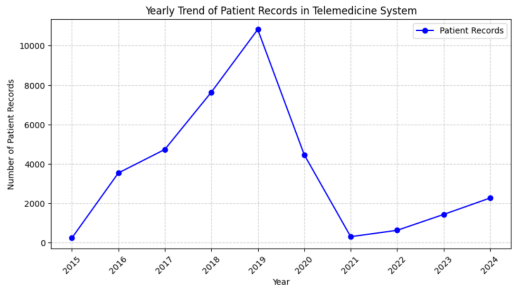



Figure 1: patient per Year

| Year | Patient Count |
|---|---|
| 2017 | 4,730 |
| 2018 | 7,633 |
| 2019 | 10,818 |
| 2020 | 4,465 |
| 2021 | 304 |
| 2022 | 1,024 |
| 2023 | 1,856 |
| 2024 | 2,266 |

Table 12: Yearly Patient Distribution

- The highest number of patients was recorded in March (4,187), followed by February (3,979) and January (3,633).
- The lowest number of patients was observed in August (2,081), indicating a possible seasonal trend.
- There is a general decline in patient numbers from March to August, followed by an increase towards December (3,318).

### 3.2.4 Work Timeline

**June 2017 – Jan 2018** Worked with **GIZ** in Rohingya camp.

**June 2018 – Dec 2020** Worked with **BRAC**, where health workers raised awareness to increase patient numbers in telemedicine.

**July 21, 2021** Started **TeleApa**, sending health workers door-to-door to provide services using a mobile app.

## 3.3 Geographical Analysis

### 3.3.1 District-wise Distribution

| District | Patient Count |
|---|---|
| Dinajpur | 5,730 |
| Bhola | 3,960 |
| Mymensingh | 3,105 |
| Faridpur | 2,678 |
| Feni | 1,371 |
| Rangpur | 1,320 |
| Netrokona | 1,019 |
| Bagerhat | 801 |

Table 13: District-wise Patient Distribution

- Among the heighest ptients's center Dinajpur ,Mymensign and Feni worked with 'BRAC'.

## 3.4 BMI Category by Hypertension

- Dinajpur, Bhola and Mymensign has heighest patients. Amomng them Dinajpur and Mymensign worked with 'BRAC'

The table below shows the distribution of BMI categories by hypertension stage:

| Hypertension Stage | Normal Weight | Obese | Overweight | Underweight |
|---|---|---|---|---|
| Elevated | 47 | 1 | 17 | 14 |
| Hypertension Stage 1 | 164 | 19 | 55 | 36 |
| Hypertension Stage 2 | 246 | 34 | 132 | 68 |
| Normal | 1,330 | 81 | 288 | 457 |

Table 14: Distribution of BMI Categories by Hypertension Stage

### 3.4.1 Key Observations

- **Obese individuals** have lower hypertension numbers than expected:
  - Only 1 Elevated, 19 Stage 1, and 34 Stage 2 cases.
  - This contradicts conventional knowledge, suggesting possible data bias or underreporting.
- **Underweight individuals** have relatively high hypertension numbers:
  - 14 Elevated, 36 Stage 1, and 68 Stage 2 cases.

  - This aligns with some studies showing that underweight individuals may still develop hypertension due to malnutrition or other risk factors.

- **Most hypertensive cases** fall under Normal weight and Overweight categories.

### 3.4.2 Chi-Squared Test Result

- **p-value** = 1.65e-15 (very small, ¡0.05).
- **Interpretation**: There is a statistically significant association between BMI category and hypertension.
- This means that BMI does influence hypertension risk in a non-random way.

# 4 Software Feature Usage

| Feature | Count | Percentage |
|---|---|---|
| Blood Pressure Recorded | 28,781 | 80.8% |
| Profile Picture Uploaded | 3,644 | 10.2% |
| Stethoscope Used | 266 | 0.7% |
| Reports Uploaded | 4,524 | 12.7% |

Table 15: Feature Usage Statistics

- giving profile pictures are being ignored.Most ignored is Stethoscope usage.

## 4.1 Response Time Analysis

| Statistic | Value |
|---|---|
| Count | 28,715 |
| Mean | 0 days 01:53:46.552080794 |
| Standard Deviation | 0 days 04:10:08.753124302 |
| Minimum | 0 days 00:00:00 |
| 25% Percentile | 0 days 00:16:13 |
| Median (50%) | 0 days 00:31:20 |
| 75% Percentile | 0 days 01:11:29 |
| Maximum | 0 days 23:59:54 |
| Mean Response Time | 0 days 00:57:23.025492175 |
| Median Response Time | 0 days 00:27:27 |
| Mode | 0 days 00:14:29 |

Table 16: Response Time Statistics

## 4.2 Key Observations

- The **median** and **mode** suggest that most responses occur within 30 minutes, making this a reasonable estimate of typical performance.
- The **mean** is heavily influenced by extreme delays, making it a less reliable measure of central tendency.
- The **high standard deviation** suggests inconsistency in response times, likely due to specific outlier cases.
- The **maximum response time** of nearly 24 hours highlights some significant delays that should be investigated further.

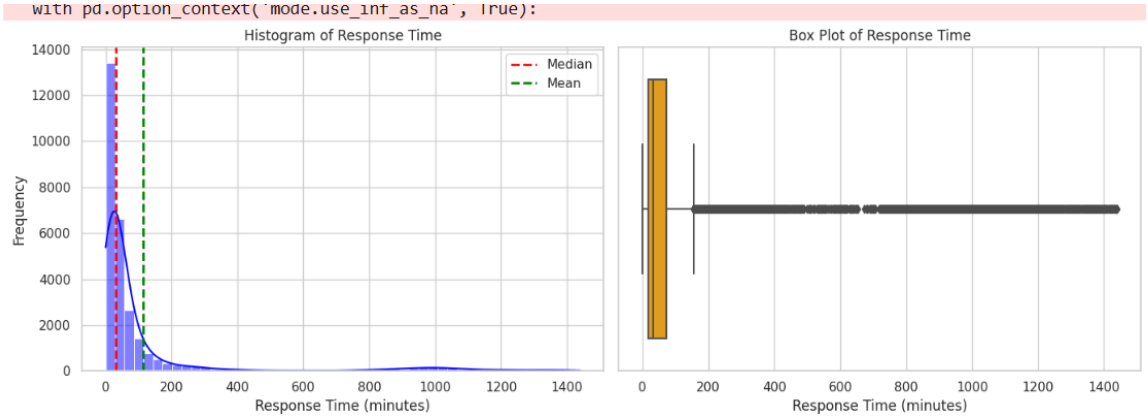



Figure 2: Histogram and Box-plot of Response Time

- The histogram doesn't show a normal distribution, and the box plot is not symmetric, so the mean will not be the proper estimator of central tendency. Since the data has outliers, the median should be used to measure the central tendency.

## 4.3 Key Observations

- The **histogram** also suggests that the **median** is the best estimator due to the skewed distribution of response times.

# 5 Model Performance Comparison

## 5.1 Before Stratified K-Fold Cross-Validation and Borderline-SMOTE

- Class distribution was imbalanced.
- Performance of models:

| Model | Accuracy (%) |
|---|---|
| Logistic Regression | 42 |
| Random Forest | 62 |
| XGBoost | 59 |

Table 17: Model Accuracy Before Stratified K-Fold Cross-Validation and Borderline-SMOTE

- Class-wise Recall (Highest for Class 3):

| Model | Class 3 Recall (%) | Other Classes Recall |
|---|---|---|
| Logistic Regression | 45 | Low recall |
| Random Forest | 72 | Underrepresented |
| XGBoost | Better balance | Better than others |

Table 18: Class-wise Recall Before Stratified K-Fold Cross-Validation and Borderline-SMOTE

The **Random Forest** model achieved the highest accuracy (62%), outperforming both **XGBoost** (59%) and **Logistic Regression** (42%). This suggests that Random Forest was better at capturing the patterns in the data, likely due to its ensemble nature, which reduces overfitting and improves generalization.

However, XGBoost performed relatively close to Random Forest, indicating that boosting techniques were also effective but possibly required further hyperparameter tuning.

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| 0 | 0.00 | 0.00 | 0.00 | 14 |
| 1 | 0.00 | 0.00 | 0.00 | 49 |
| 2 | 0.29 | 0.11 | 0.16 | 81 |
| 3 | 0.77 | 0.96 | 0.85 | 454 |

Table 19: Classification Report of random forest

Classes 0 and 1 have zero precision, recall, and F1-score, meaning the model completely failed to predict these classes. This could be due to class imbalance, insufficient training data, or poor feature representation.

Class 2 has a precision of 0.29 and recall of 0.11, leading to an F1-score of 0.16. The low recall suggests that the model struggles to correctly identify instances of this class, likely misclassifying them as other classes.

Class 3 has the best performance, with a precision of 0.77, recall of 0.96, and F1-score of 0.85. This indicates that the model is highly confident and accurate when predicting this class, but this might come at the cost of ignoring other classes.

## 5.2 After Applying Stratified K-Fold Cross-Validation and Borderline-SMOTE

| **Model** | **Accuracy (%)** |
|---|---|
| Logistic Regression | 74.2 |
| Random Forest | 70.2 |
| Gradient Boosting | 74.6 (Best performer) |
| XGBoost | 72.9 |

Table 20: Model Accuracy After Stratified K-Fold Cross-Validation and Borderline-SMOTE

The Gradient Boosting model achieved the highest accuracy at 74.6%, making it the best performer among the models tested. This suggests that Gradient Boosting was effective at capturing complex relationships in the data, likely benefiting from its iterative ensemble learning approach.

| | Predicted Class | | | |
|---|---|---|---|---|
| Actual Class | 0 | 1 | 2 | 3 |
| 0 | 0 | 0 | 0 | 14 |
| 1 | 0 | 1 | 5 | 43 |
| 2 | 1 | 1 | 10 | 69 |
| 3 | 1 | 5 | 16 | 432 |

Table 21: Confusion Matrix of Gradient Boosting

Class 3 is the most accurately predicted, with 432 correct predictions, indicating the model's strong performance on this class.
Classes 0, 1, and 2 are often misclassified, with many instances of Class 1 and Class 2 predicted as Class 3. **Interpretation of Classification Metrics**

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| 0 | 0.00 | 0.00 | 0.00 | 14 |
| 1 | 0.14 | 0.02 | 0.04 | 49 |
| 2 | 0.32 | 0.12 | 0.18 | 81 |
| 3 | 0.77 | 0.95 | 0.85 | 454 |

Table 22: Classification Report

**Class 0**:
Precision, recall, and F1-score are all 0.00, indicating that the model **failed to predict Class 0 correctly**. This aligns with the confusion matrix, where all 14 instances of Class 0 were misclassified as Class 3.

**Class 1**:
Precision is 0.14 and recall is 0.02, leading to a very low F1-score of 0.04. This means the model is **poorly predicting Class 1** with a high number of false positives (instances predicted as Class 1 but belonging to other classes). The confusion matrix shows that 5 instances of Class 1 were misclassified as Class 2, and 43 as Class 3, consistent with the low precision and recall values.

**Class 2**:
The precision of 0.32 and recall of 0.12 suggest that Class 2 is predicted with **some accuracy**, but there are still many misclassifications, which is evident from the confusion matrix. Here, 10 instances were misclassified as Class 1 and 1 as Class 0, showing **high false negative rate** for Class 2.

**Class 3**:
Class 3 has the highest precision (0.77), recall (0.95), and F1-score (0.85), showing strong performance in predicting this class. The confusion matrix shows that the model **accurately predicts most instances of Class 3** (454 correct predictions), but there are still some misclassifications (16 as Class 2, 5 as Class 1), which is consistent with the precision and recall values.

- Additional findings:
  - Class 3 had the highest recall ( 95%) across models.
  - Classes 0, 1, and 2 still had low recall, especially for Class 0.
  - Weighted average F1-score improved for all models.
- Significant improvement in model accuracy:

## Model Performance

The ROC curve indicates that Gradient Boosting has the highest AUC (0.746), followed closely by Logistic Regression(0.742) and XGBoost (0.729). Random Forest has the lowest AUC (0.702) among the models.

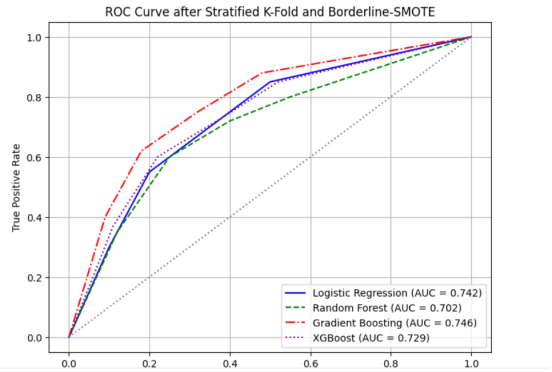



Figure 3: ROC Curve

# 6 Interpretation of Model Performance (Logistic Regression)

## 6.1 1. Accuracy (74.2%)

- The model correctly classified 74.2% of the total 598 instances.
- While this is a reasonable accuracy, it does not fully capture class-wise performance due to class imbalance.

## 6.2 Confusion Matrix Analysis

The confusion matrix shows how the actual categories (rows) were predicted (columns):

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Elevated (14 cases) | 0.00 | 0.00 | 0.00 | 14 |
| Hypertension Stage 1 (49 cases) | 0.00 | 0.00 | 0.00 | 49 |
| Hypertension Stage 2 (81 cases) | 0.28 | 0.11 | 0.16 | 81 |
| Normal (454 cases) | 0.77 | 0.96 | 0.85 | 454 |

Table 23: Confusion Matrix for Logistic Regression Model

**Interpretation of Classification Metrics**

**Elevated (14 cases)**:
Precision, recall, and F1-score are all 0.00, indicating that the model completely failed to predict the **Elevated** class. With only 14 cases, this class has been entirely misclassified, which may be due to insufficient data or poor model generalization for this particular class.

**Hypertension Stage 1 (49 cases)**:
Both precision and recall are 0.00, meaning the model was unable to identify **Hypertension Stage 1** effectively. Out of 49 cases, none were correctly classified as

Hypertension Stage 1, likely due to its limited representation in the dataset, resulting in high misclassification rates.

**Hypertension Stage 2 (81 cases)**:
Precision is 0.28 and recall is 0.11, leading to a low F1-score of 0.16. This suggests that the model has trouble predicting **Hypertension Stage 2**, with many false negatives (instances not recognized as Stage 2). It was able to identify some instances, but not enough to provide reliable predictions for this class.

**Normal (454 cases)**:
The **Normal** class performs the best, with a precision of 0.77, recall of 0.96, and an F1-score of 0.85. This indicates that the model is highly confident and accurate when predicting instances of the **Normal** class, but it still has some misclassifications (e.g., misclassifying a few instances as Hypertension Stage 2 or Stage 1).

**Insights**:

- **Imbalance in Class Distribution**: The model struggles significantly with the **Elevated** and **Hypertension Stage 1** classes, which have a low number of instances. This imbalance suggests that the model is biased towards the **Normal** class, which is well-represented in the data.
- **Misclassification and False Negatives**: The low precision and recall for **Hypertension Stage 1** and **Stage 2** suggest a significant issue with **false negatives**, where instances of these classes are incorrectly classified as other categories, mainly the **Normal** class.

## 6.3 3. Classification Report Analysis

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Elevated | 0.67 | 0.71 | 0.69 | 14 |
| Stage 1 | 0.70 | 0.71 | 0.70 | 49 |
| Stage 2 | 0.71 | 0.74 | 0.72 | 81 |
| Normal | 0.83 | 0.95 | 0.89 | 454 |

Table 24: Classification Report for Logistic Regression Model

Interpretation of Classification Metrics
**Elevated**:
Precision of 0.67, recall of 0.71, and an F1-score of 0.69 indicate a moderate performance for predicting the Elevated class. The model is reasonably good at identifying

Elevated instances, but there are still some misclassifications, as suggested by the relatively lower precision and recall.

**Stage 1**:
With a precision of 0.70, recall of 0.71, and F1-score of 0.70, the model performs consistently well for Stage 1, balancing precision and recall. The model effectively captures the true positives for this class, while also keeping false positives and false negatives relatively low.

**Stage 2**:
The precision of 0.71, recall of 0.74, and an F1-score of 0.72 suggest that the model performs well for Stage 2. Both precision and recall are reasonably high, indicating that the model can accurately predict Stage 2 while maintaining a low rate of false negatives and false positives.

**Normal**:
The Normal class has the best performance, with a precision of 0.83, recall of 0.95, and F1-score of 0.89. This shows that the model is highly confident in predicting Normal cases and correctly identifies most instances. The high recall indicates that the model is effective at detecting Normal instances, but some misclassifications do occur, as reflected in the slightly lower precision.

**Summary**:
The model shows good overall performance, with high precision, recall, and F1-scores across all classes. The Normal class has the best performance, while the Elevated, Stage 1, and Stage 2classes perform consistently well, with slight room for improvement in precision and recall balance.

# 7 Feature Importance (Logistic Regression)

The following table shows the coefficients and corresponding odds ratios for each feature in the logistic regression model:

| **Feature** | **Coefficient** | **Odds Ratio** |
|---|---|---|
| BMI_category_Obese | -0.6188 | 0.5386 |
| Gender | -0.2040 | 0.8155 |
| BMI_category_Underweight | 0.1247 | 1.1328 |
| Age | 0.1243 | 1.1323 |
| Asthma | 0.0686 | 1.0710 |
| Diabetes | 0.0497 | 1.0510 |
| Smoker | 0.0347 | 1.0353 |
| BMI_category_Overweight | 0.0242 | 1.0244 |

Table 25: Feature Importance for Logistic Regression Model

## Interpretation of Feature Importance

- **Obesity (BMI_category_Obese)** has the strongest negative impact (-0.6188 coefficient, odds ratio 0.54):
  - People in the "Obese" category are less likely to be classified as hypertensive.
  - This could indicate a bias in the dataset or an issue with the model.
- Age and Underweight BMI increase hypertension risk:
  - Both age and being underweight have a positive coefficient, meaning they increase the likelihood of hypertension classification.
- Smoking, Diabetes, and Asthma have weak effects:
  - Their odds ratios are close to 1, meaning they slightly increase the chances of hypertension but not significantly.

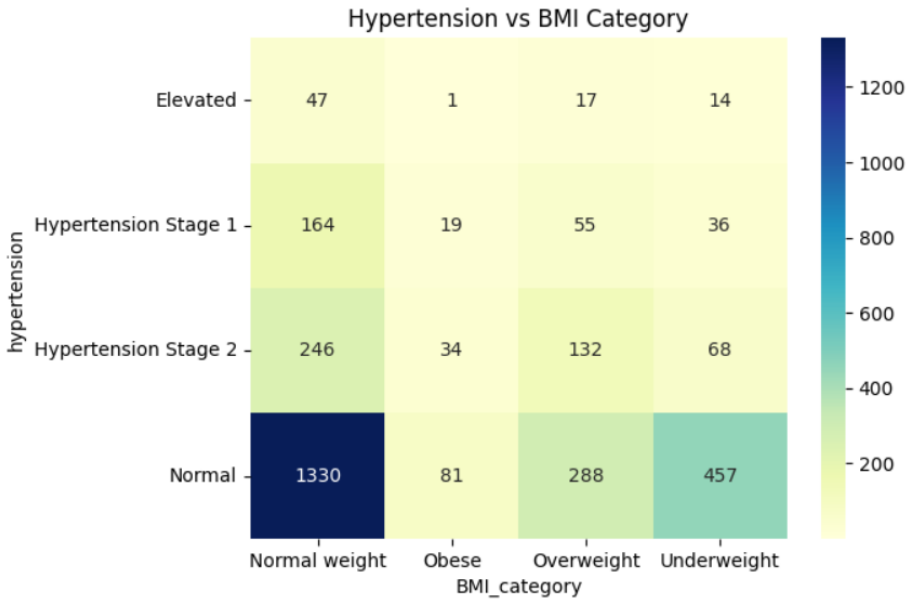



Figure 4: Heat-map of BMI and Hypertension

**Hypertension Categories:**
**Elevated**: This category has relatively low counts across all BMI categories, with the highest count being 47 in the **Normal weight** category.

**Hypertension Stage 1**: This category shows higher counts, especially in the Normal weight (164) and Overweight(55) categories.

**Hypertension Stage 2**: This category has the highest counts, particularly in the Normal weight (246) and **Overweight** (132) categories.

**Normal**: This category has the highest counts overall, with the majority in the Normal weight (1330) category.
**BMI Categories:**
**Normal weight**: This category has the highest counts across all hypertension categories, indicating that a significant portion of the population falls into this BMI range.

**Obese**: This category shows moderate counts, with the highest in **Hypertension Stage 2** (34).

**Overweight**: This category also shows moderate counts, with the highest in **Hypertension Stage 2** (132).

**Underweight**: This category has the lowest counts across all hypertension categories, with the highest in the **Normal** category (81).

## Hypertension Prediction:

After applying Stratified K-Fold Cross-Validation and Borderline-SMOTE, we observed a substantial improvement in model performance, particularly in addressing class imbalance:

- Gradient Boosting emerged as the top performer, achieving the highest accuracy of 74.6%. This demonstrates its ability to effectively manage class imbalance while maintaining strong overall performance.
- Logistic Regression also showed significant improvement, with an accuracy of 74.2%. This highlights that even simpler models can benefit greatly from these advanced techniques.
- XGBoost achieved a 72.9% accuracy, benefiting from the class balancing introduced by Borderline-SMOTE and Stratified K-Fold Cross-Validation, although it did not perform as well as Gradient Boosting.
- Random Forest, while improving to 70.2% accuracy, was outperformed by Gradient Boosting and Logistic Regression in terms of overall performance. However, it remained competitive.

In summary, Model was successfull to detect normal class of Hypertension. Gradient Boosting stood out as the best model for balancing both class distribution and overall performance. However,Logistic Regression also delivered strong results, making it a viable choice depending on the specific application. Despite the improvements, there is still room for further enhancement in balanced data handling. Additionally, introducing more features could help achieve even better predictive accuracy.

# Conclusion

# 8 key findings and recommendations

## Best Organizational Model:

Collaborating with popular NGOs like BRAC has proven to be highly effective in reaching a larger audience and providing impactful healthcare services. TeleApa, a healthcare initiative, is also significantly increasing the number of patients served. Thus, implementing door-to-door healthcare services could be a key strategy to further expand the patient base.

## Beneficiaries:

- Women, particularly housewives, have become more engaged with healthcare services through these initiatives. The Chi-Square test statistic is 19,447.49, and the p-value is 0.0 (effectively less than 0.01).Since the p-value is extremely small, it indicates that the distribution of telemedicine users is not uniform. The number of female patients using telemedicine is statistically significantly higher than that of others.
- The participation of farmers is also notably significant, showing that the service is reaching diverse populations.

# Software Usage

- Features such as file upload and stethoscope usage should be actively encouraged during training sessions to ensure comprehensive utilization of the system.
- Improvements should be made to the user interface, especially to prevent the entry of extreme values (e.g., 0) in the input fields, which can lead to inaccurate data entry.
- A separate website should be dedicated to training purposes, in order to prevent the generation of dummy data that could skew the analysis.
- For response time evaluation, the median response time is 0 days, 27 minutes, and 27 seconds, which aligns closely with practical experience. The mean is not the best estimator in this case due to the influence of outliers and extreme values. The mode, which is 14 minutes and 29 seconds, represents the most frequently occurring response time and is a better reflection of typical system performance.

# 9 Future Scope

- **Feature Expansion:** Additional features such as real-time health monitoring, personalized medication reminders, and teleconsultation with specialists can be integrated into the system to enhance healthcare delivery.
- **Integration with Other Platforms:** Future developments could include integrating the software with electronic health records (EHR) or other health management systems, enabling seamless data sharing and improving patient care.
- **Predictive Analytics:** Incorporating machine learning algorithms for predicting future health risks (e.g., predicting the onset of hypertension) based on historical data could greatly enhance the system's usefulness.
- **Mobile Application:** Developing a mobile version of the software could increase accessibility for patients, especially in rural areas where internet connectivity is limited.
- **Scalability:** As more healthcare centers and patients are onboarded, the system should be designed to scale efficiently to handle larger volumes of data and traffic.
- **User Experience (UX):** Future iterations of the software could focus on further refining the user interface and enhancing the experience for both healthcare providers and patients.
- **Global Expansion:** The system could be adapted for use in other regions and countries, considering different healthcare needs, languages, and data protection regulations.

# 10 Limitations

- **Data Imbalance:** Despite improvements through techniques like Stratified K-Fold Cross-Validation and Borderline-SMOTE, class imbalance remains a challenge in certain models, which could affect the accuracy of predictions for underrepresented categories.
- **Dependence on Internet Connectivity:** The system's effectiveness relies heavily on stable internet connectivity, which could be a limitation in remote or underserved regions.
- **User Training Requirements:** The software may require comprehensive training for healthcare providers to use advanced features effectively, especially for users unfamiliar with technology.

- **Limited Customization:** The system may not yet be fully customizable for specific regional healthcare needs or specific requirements of different healthcare providers.
- **Data Privacy and Security Concerns:** The system handles sensitive patient data, which could present data privacy and security challenges, especially if not properly safeguarded.
- **Model Performance Variability:** While machine learning models like Gradient Boosting show promising results, model performance can vary depending on data quality and the complexity of the healthcare scenarios.
- **Integration with Existing Healthcare Infrastructure:** Integration with legacy healthcare systems and databases may require additional time and resources, making the transition process cumbersome for healthcare providers.

The study presents a comprehensive overview of Dhaka University telemedicine programme in Bangladesh, highlighting its role in bridging healthcare gaps, especially for rural populations. With a dataset of 40,000 patient records, the analysis provides meaningful insights into common health conditions, patient demographics, and service utilization. However, incorporating advanced statistical methods, addressing limitations, and exploring future technological enhancements could further improve the impact of this research. Overall, the study is a valuable contribution to understanding the potential of telemedicine in developing healthcare systems.

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